



Where do Gradient Methods Really Happen? Maybe Not Where We Expected

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Intro

Modern neural networks are trained in extremely high-dimensional parameter spaces, yet their dynamics often look partly low-dimensional. Recent work observes that gradients may quickly align with a dominant Hessian subspace after only a few training steps.

The goal of our draft is to separate descriptive geometry from causal optimization mechanisms across architectures, optimizers, and alternative choices of training subspaces.

Problem

Let u_t be the update produced by an optimizer from the current gradient and optimizer state. For a subspace U_t , we decompose this update into projected and complementary parts:

$$u_t = P_{U_t} u_t + (I - P_{U_t}) u_t.$$

We compare three trajectories from the same initialization: full updates, updates restricted to U_t , and updates restricted to the orthogonal complement. The central question is causal: does a high-alignment subspace preserve loss decrease and validation quality, or does the complementary bulk component still carry the useful progress?

Projectors

The poster will compare several ways to choose U_t :

- top Hessian eigenspaces, following the dominant-subspace hypothesis;
- top Layerwise Momentum subspace selected by the cumulative energy of singular values
- top Adaptive Hessian eigenspace
- top Muon (Kronecker) Hessian eigenspace
- learnable Projector basis on the Stiefel Manifold

Metrics

We measure subspace alignment by

$$\chi_U(v) = \|P_U v\| / \|v\|.$$

Following the reference setup, an exponential moving average of alignment can be used as a switching criterion.

We also track training loss, update norms, and a projected descent ratio: how much predicted one-step loss decrease remains after projection:

$$\rho_U(v) = \frac{\eta \langle g, P_U v \rangle - \frac{\eta^2}{2} (P_U v)^\top H (P_U v)}{\eta \langle g, v \rangle - \frac{\eta^2}{2} v^\top H v}.$$

These metrics distinguish geometric overlap from actual optimization and generalization behavior.

Optimizers

We treat the optimizer as part of the geometry, not as a detail outside the projection experiment.

- SGD and SGD with momentum;
- Adam and AdamW;
- Muon and related matrix-aware updates;

Explanation

- Adaptive Hessian Subspace
The Adam preconditioner induces a local diagonal metric

$$P = \text{diag}(\sqrt{\hat{v}} + \varepsilon).$$

We therefore analyze curvature in Adam-whitened coordinates:

$$\tilde{H} = P^{-1/2} H P^{-1/2}.$$

For top eigenvectors Q , the dominant parameter-space subspace is

$$S_{\text{adaptive}} = P^{-1/2} \text{span}(Q),$$

and the update is split by the P -orthogonal projector

$$\Pi_P = P^{-1/2} Q Q^\top P^{1/2}, \quad u_{\text{dom}} = \Pi_P u, \quad u_{\text{bulk}} = (I - \Pi_P) u.$$

Thus the dominant and bulk components are orthogonal in the optimizer's adaptive metric, not necessarily in the Euclidean metric.

- Muon (Kronecker) Hessian
The Muon polar update can be represented locally by a two-sided Kronecker preconditioner:

$$\mathcal{M} = L \otimes R, \quad L = (G G^\top)^{1/2}, \quad R = (G^\top G)^{1/2}.$$

The corresponding whitened Hessian block is

$$\tilde{H}_\ell[D] = L^{-1/2} H_\ell [L^{-1/2} D R^{-1/2}] R^{-1/2}.$$

We extract dominant left/right subspaces $(\tilde{U}, \tilde{V})$ by HOOI, yielding the structured subspace

$$S_{\text{Muon}} = L^{-1/2} \text{span}(\tilde{U}) \otimes R^{-1/2} \text{span}(\tilde{V}).$$

The dom component is obtained by projection:

$$D_{\text{dom}} = L^{-1/2} \tilde{U} \tilde{U}^\top (L^{1/2} D R^{1/2}) \tilde{V} \tilde{V}^\top R^{-1/2}.$$

Theoretical results

SGD alignment does not imply projected trainability. On a quadratic model

$$L = \frac{1}{2} x^\top \Lambda x + \frac{1}{2} y^\top M y, \quad \lambda(\Lambda) \gg \mu(M),$$

SGD noise amplifies gradient energy in high-curvature directions. In the isotropic case,

$$\bar{\chi}_k = \frac{\mathbb{E} \|P_k \nabla L\|^2}{\mathbb{E} \|\nabla L\|^2} \approx \frac{k\lambda}{k\lambda + (d-k)\mu},$$

so the gradient can appear almost fully aligned with the dominant Hessian subspace when

$$\lambda/\mu \gg (d-k)/k.$$

But if we actually train only in that subspace,

$$\dot{\theta} = -P_k \nabla L(\theta),$$

then the bulk variables are frozen:

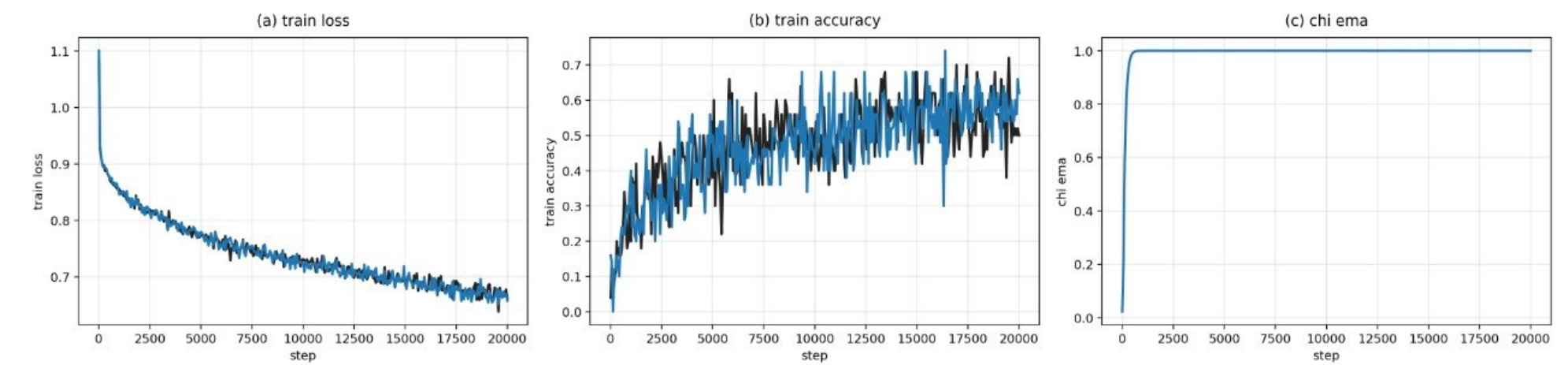
$$\dot{y} = 0.$$

Projected training therefore stops on

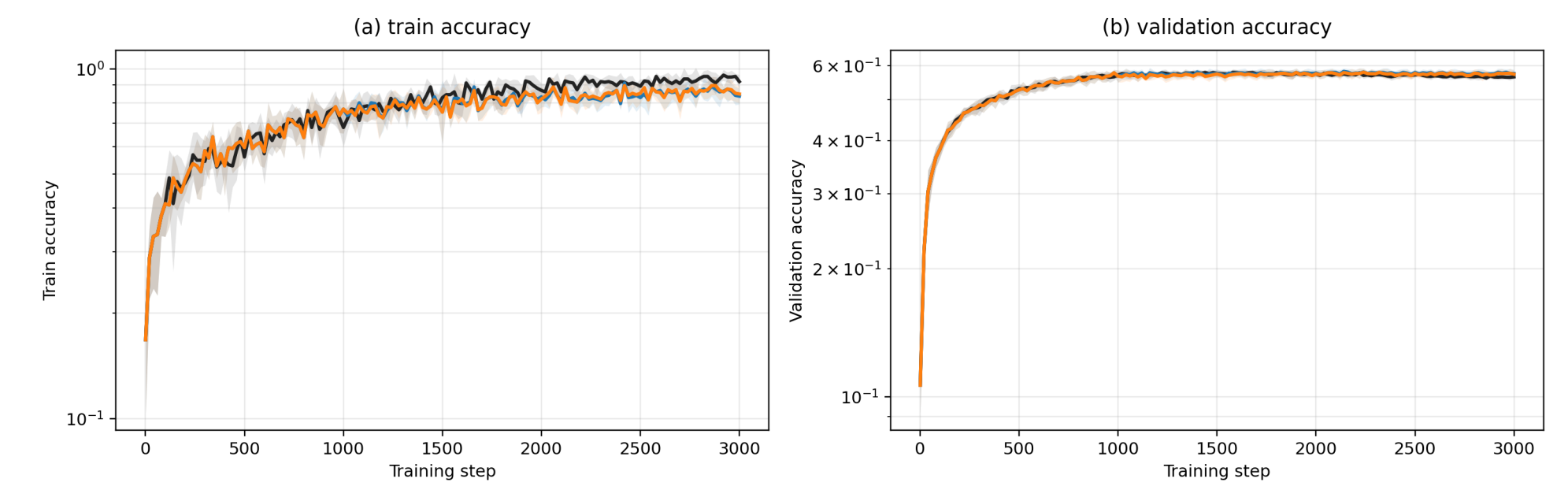
$$\theta^* + U_\perp y$$

rather than at the true optimum, leaving residual loss whenever $y \neq 0$.

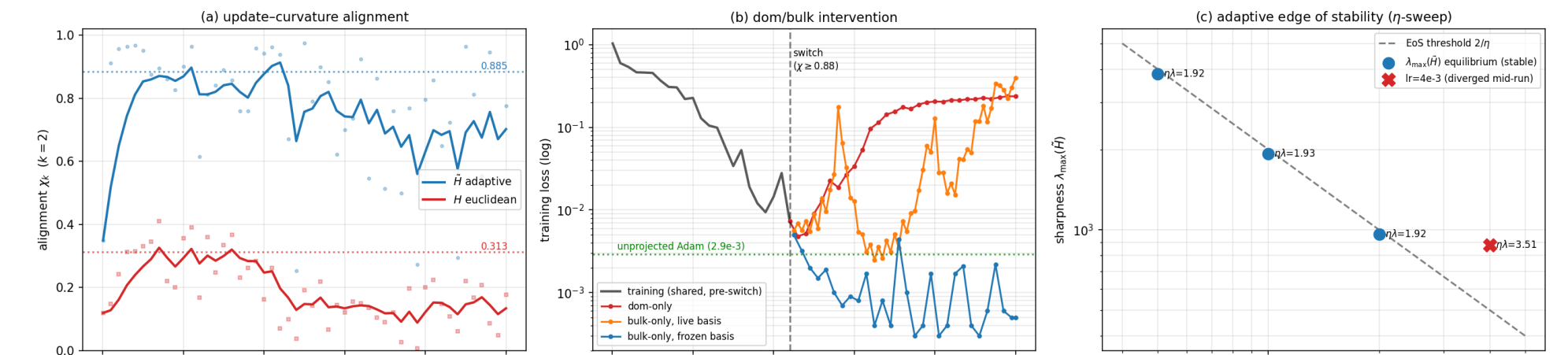
Experiments



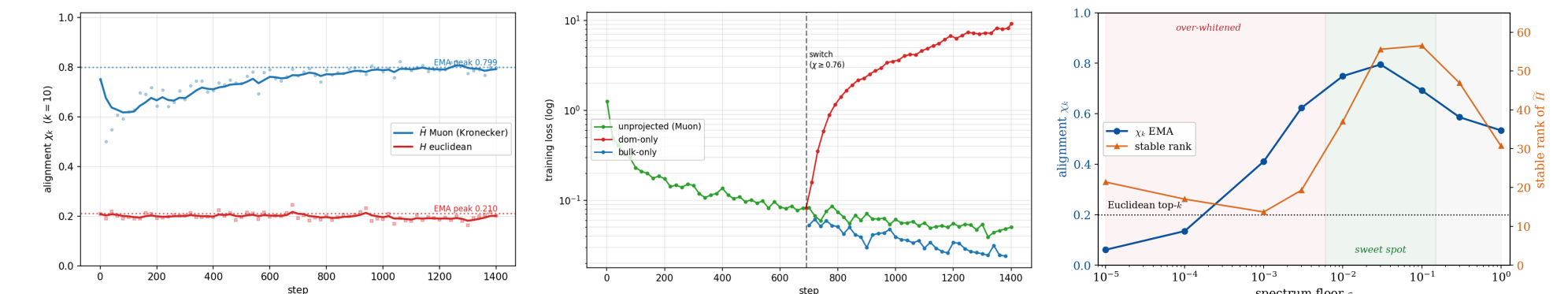
Projected-training comparison for SGD with a learnable Stiefel projector on CNN on CIFAR10.



Projected-training norm correction for AdamW with Layerwise Momentum projector. Norm correction has almost no visible effect on the behavior of the momentum-induced projection.



The update aligns with the top-k subspace of the Adaptive Hessian. Projecting onto that subspace halts training, meanwhile the top eigenvalue of the Adaptive Hessian stays pinned to the EoS threshold.



The update lives in the top-k Muon (Kronecker) Hessian subspace, and projecting onto it makes Muon diverge while the bulk carries the descent below unprojected Muon. The Muon-metric whitening is sharpest at an intermediate spectrum.

Acknowledgements

This research was supported in part through computational resources of HPC facilities at HSE University, see Kostenetskiy et al. [1].

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